1. **Create a network with a minimum of 5 devices using various network topologies using Cisco Packet Tracer tool and test the connectivity with the transmission of a simple PDU.(Intermediate Device – switch).**

Ans.

1. Selecting Devices and a Topology

You'll need:

- **1 Switch** (intermediate device)
- **4 PCs** (end devices, you can add more if desired)
- **Cabling** (Automatic/straight-through for PC-to-switch)

Example Topologies:

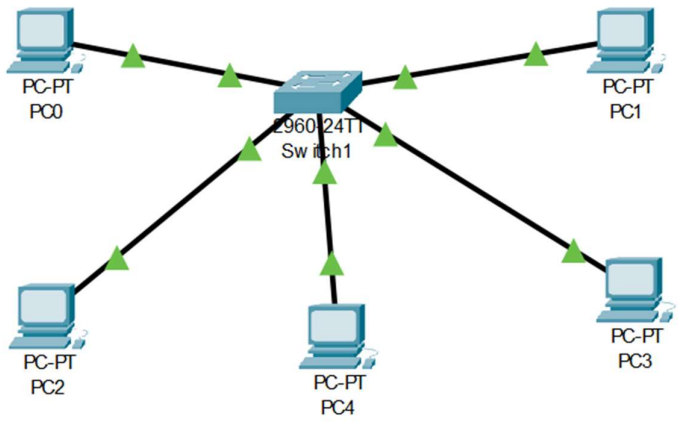
- **Star Topology:** All PCs connected to a central switch.
- **Bus Topology:** Devices connected in a line via a switch or using a single cable (simulated with Packet Tracer).
- **Ring Topology:** Devices connected around a loop (not typical with real switches but can be emulated).

Here we'll use a simple star topology for maximum clarity; each PC is directly attached to the switch.

2. Building the Network

a. Place Devices:

- Drag and drop one **Switch** and five **PCs** onto the workspace.



b. Connect Devices:

- Use the **Automatic Connection Tool** (lightning bolt icon) or manually use straight-through cables.

- Connect each PC's FastEthernet port (e.g., FastEthernet0) to an available switch port (e.g., FastEthernet0/1, 0/2, ...).

3. Assign IP Addresses

For example (Class C addressing):

Device	IP Address	Subnet Mask
PC0	192.168.1.2	255.255.255.0
PC1	192.168.1.3	255.255.255.0
PC2	192.168.1.4	255.255.255.0
PC3	192.168.1.5	255.255.255.0
PC4	192.168.1.6	255.255.255.0

To set IP on a PC:

- Click PC → Desktop tab → IP Configuration → Enter IP Address and Subnet Mask.

4. Test Connectivity with a Simple PDU

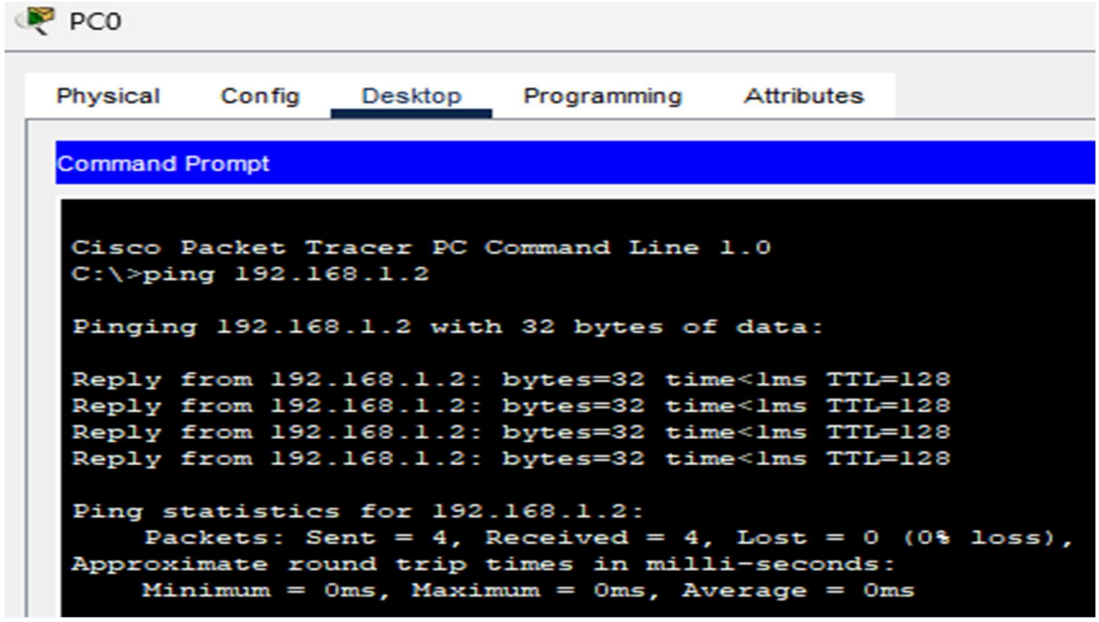
Send a Simple PDU:

- Click the **Add Simple PDU** icon .
- Click the source PC (e.g., PC3), then the destination PC (e.g., PC0).

You will see an envelope icon next to the source and destination. Successful delivery will show as **green**; if not, it will be **red**.

Alternative: Ping via CLI

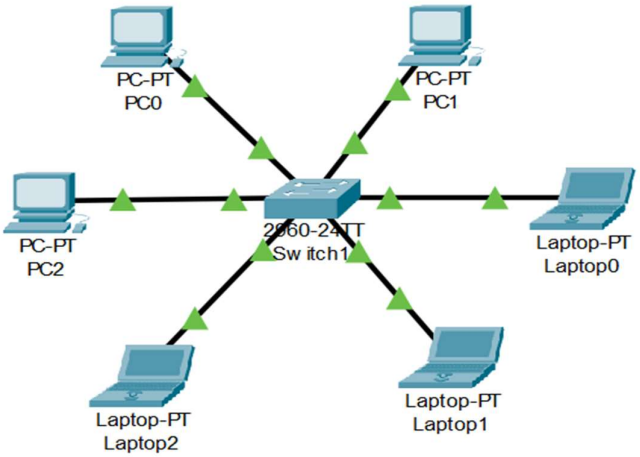
- Open PC3 → Desktop → Command Prompt.
- ping 192.168.1.2. Replies indicate successful connectivity.



5. What Does the Switch Do?

- The switch **learns each PC's MAC address** and **forwards traffic only to the correct port**, ensuring efficient communication.

2. Create a network with a minimum of 5 devices using various network topologies using Cisco Packet Tracer tool and test the connectivity with the transmission of a simple PDU. (Intermediate Device – switch).



Star Topology with Six End Devices Using a Switch in Cisco Packet Tracer

Step-by-Step Implementation

1. Building the Star Topology

- **Devices Needed:** 1 Switch, 3 PCs, 3 Laptops, and appropriate cables.
- **Procedure:**
    - Drag and drop a switch (e.g., 2960) and six end devices (3 Laptops, 3 PCs) onto the workspace.
    - Use automatic or straight-through cables to **connect each device's FastEthernet port to a port on the switch**.
    - Result: All devices are centrally connected via the switch, forming a star topology.

2. Assigning IP Addresses

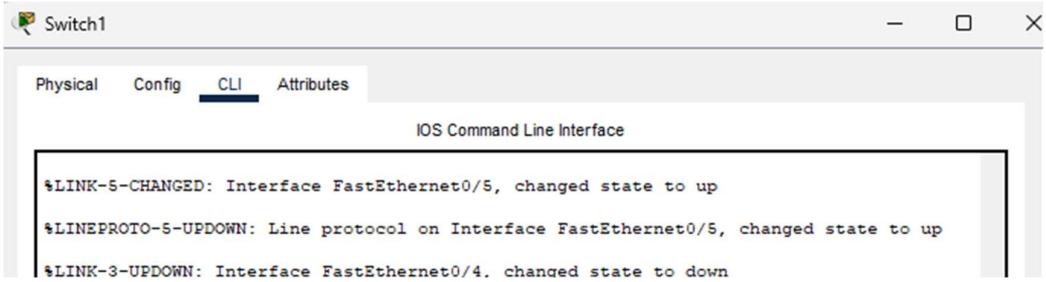
- **Example IP Scheme:**
    - PC1: 192.168.1.2
    - PC2: 192.168.1.3
    - PC3: 192.168.1.4
    - Laptop1: 192.168.1.5
    - Laptop2: 192.168.1.6
    - Laptop3: 192.168.1.7
    - Subnet Mask: 255.255.255.0 (all)
- Configure on each device via: Desktop > IP Configuration3.

3. Simulate Data Transmission (Simple PDU)

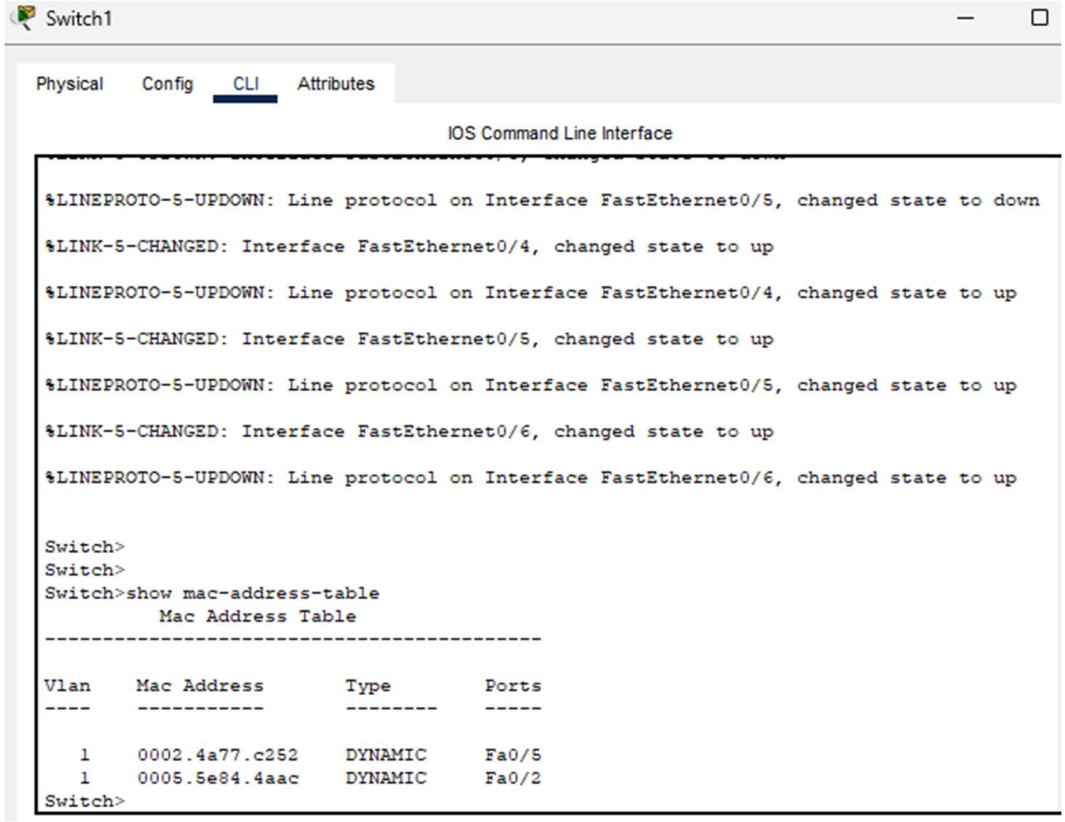
- **Sending from Device #1 to Device #4 (e.g., PC1 to Laptop1):**
    - Click the "Add Simple PDU" tool (envelope icon).
    - Click the source device, then the destination device.
    - Observe the animated path. If green, the message was successful. Alternatively, use ping in the command prompt: ping 192.168.1.5 (from PC1 to Laptop1).

4. Checking the MAC Address Table

- After transmitting packets, the switch learns the source MAC addresses.
- **To view the table:**
    - Click the switch > go to CLI > enter show mac address-table



- You'll see the list of MACs and which port they're associated with.



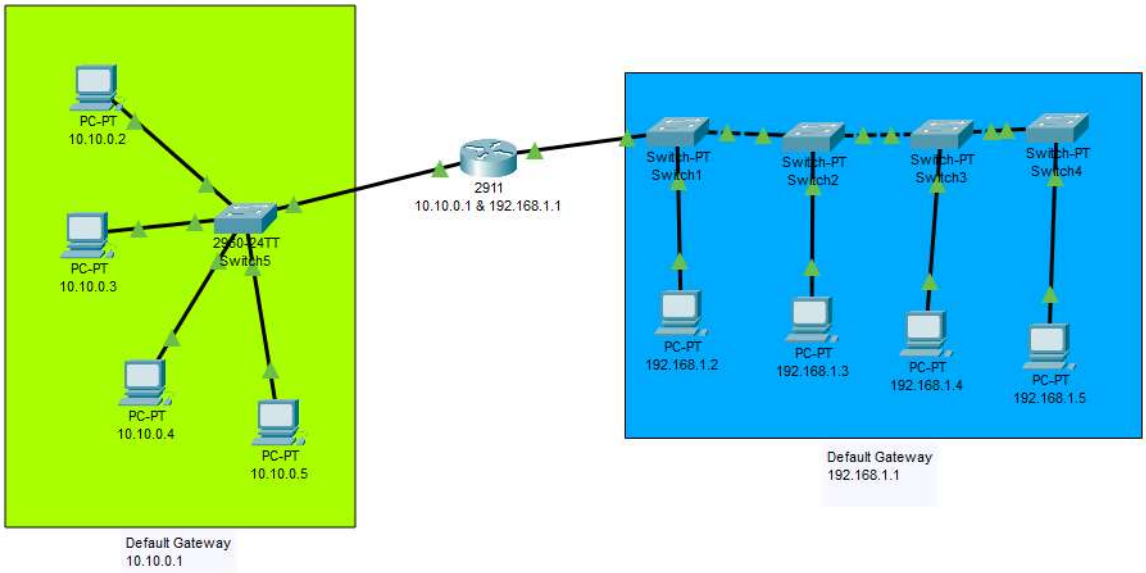
Hub vs Switch: Comparison

Feature	Hub	Switch
Layer	Physical (Layer 1)	Data Link (Layer 2)
Data Handling	Broadcasts to all ports	Sends only to destination port (unicast, multicast)
MAC Address Table	Not maintained	Maintains dynamic MAC address table
Device Intelligence	Non-intelligent	Intelligent, learns device MAC addresses
Collisions	One large collision domain (all devices)	Each port is a separate collision domain

Feature	Hub	Switch
Speed & Bandwidth	Shared bandwidth, usually slower (10Mbps)	Dedicated port bandwidth, up to 1Gbps or more
Security & Efficiency	Less secure, more network congestion	More secure, less congestion
Use Today	Largely obsolete	Standard in modern networks

3. **Implement two LANs is using any two different topologies. Set up a router to interconnect LAN. Simulate the transmission of a simple PDU (packet or message).**

Ans.



1. Network Overview

You have designed **two LANs**:

- **LAN 1 (Left side, green):** IP range 10.10.0.x using Switch5.
- **LAN 2 (Right side, blue):** IP range 192.168.1.x using Switch1, Switch2, Switch3, and Switch4.

These two LANs are interconnected with a router (2911), facilitating communication between the two subnets.

2. Device Placement and Physical Topology

A. LAN 1 (Green Section: Star Topology)

- **Devices:** 4 PCs (PC-PTs) connected to Switch5 (2960-24TT).
- **Switch5** is centrally placed, making this a Star Topology.
- **Connections:** Each PC connects directly to Switch5. Switch5 connects to the router.

B. LAN 2 (Blue Section: Extended Star Topology)

- **Devices:** 5 PCs, each connected to their own switch (Switch1–Switch4).
- **Each switch** is uplinked, most likely cascading or forming an extended star via inter-switch links.
- **All 5 PCs** connect through their respective switches and then to the router via an uplink.

C. Router (2911)

- **Interfaces:**
  - One interface for the 10.10.0.x network (LAN 1): 10.10.0.1
  - Another for the 192.168.1.x network (LAN 2): 192.168.1.1

3. IP Address Table

LAN 1 (Green, 10.10.0.x)

Device	IP Address	Subnet Mask	Default Gateway
PC-PT 1	10.10.0.2	255.0.0.0	10.10.0.1
PC-PT 2	10.10.0.3	255.0.0.0	10.10.0.1
PC-PT 3	10.10.0.4	255.0.0.0	10.10.0.1
PC-PT 4	10.10.0.5	255.0.0.0	10.10.0.1

LAN 2 (Blue, 192.168.1.x)

Device	IP Address	Subnet Mask	Default Gateway
PC-PT 5	192.168.1.2	255.255.255.0	192.168.1.1
PC-PT 6	192.168.1.3	255.255.255.0	192.168.1.1
PC-PT 7	192.168.1.4	255.255.255.0	192.168.1.1
PC-PT 8	192.168.1.5	255.255.255.0	192.168.1.1

Router (2911)

- Interface to LAN 1: 10.10.0.1/24
- Interface to LAN 2: 192.168.1.1/24

4. Step-by-Step Configuration

Step 1: Place All Devices

- Insert 1 router (2911), 5 switches (1 for LAN 1, 4 for LAN 2), and 8 PCs as shown.

Step 2: Connect the Devices

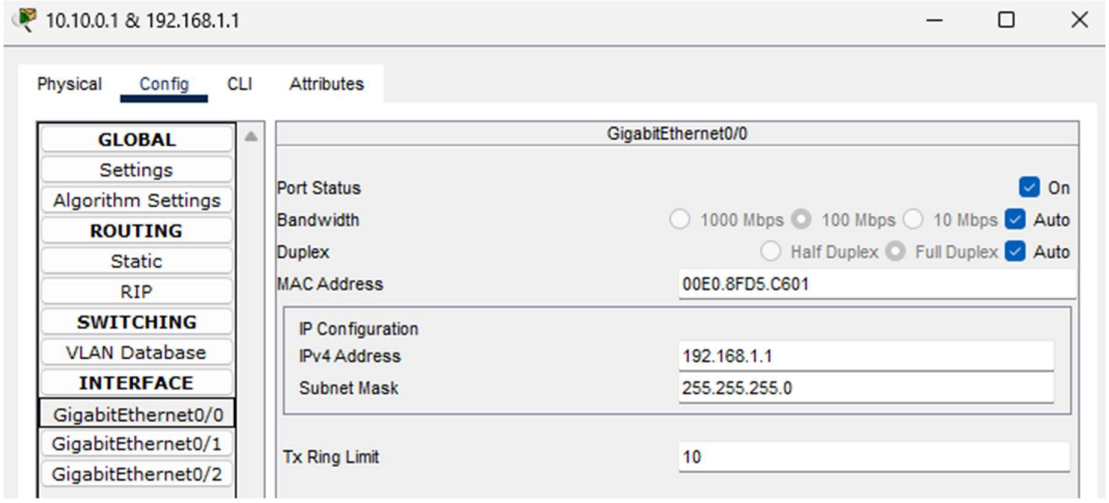
- **LAN 1:** Connect all 4 PCs to Switch5.
- **LAN 2:** Each PC connects to its own switch (Switch1–Switch4), ensuring clear paths for simulation and easier troubleshooting.
- **Switches to Router:** Use appropriate Ethernet cables to connect Switch5 to 2911's G0/0, Switch1/Switch2/Switch3/Switch4 to the router's G0/1 (via uplink ports or a central connection point).

Step 3: Configure IP Addresses

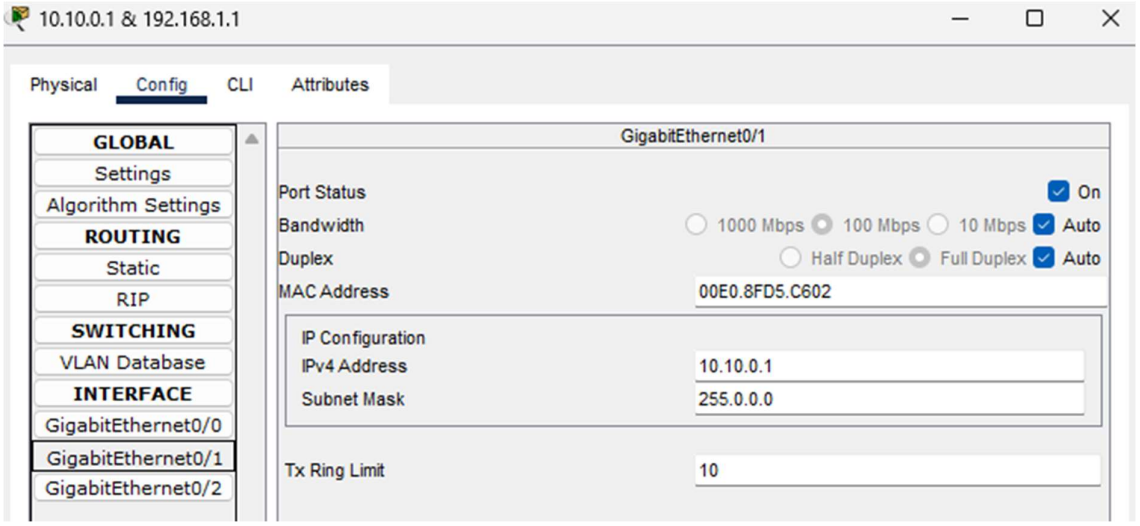
For each PC:

- Go to **Desktop > IP Configuration**.
- Set the IP address, subnet mask, and **default gateway** as shown in Step 3.

Step 4: Configure the Router



- Goto router then select **Gi0/0** then configure it as shown in the above figure.



- Goto router then select **Gi0/1** then configure it as shown in the above figure.
- Confirm with show ip interface brief that both interfaces are up and assigned their correct IPs.

Step 5: Test Communications

- On each PC, open the **Command Prompt** and ping the relevant default gateway:
  - For LAN 1: ping 10.10.0.1
  - For LAN 2: ping 192.168.1.1
- Then test cross-LAN communications (e.g., PC-PT 1 ping PC-PT 5).

Step 6: Simulate a Simple PDU Transmission

- Click the **Add Simple PDU** tool (envelope icon).
- Click a source PC in LAN 1 and a destination PC in LAN 2 (or vice versa).
- Observe the path; successful PDU transmissions will display green indicators.

5. Explanation and Topology Choices

- **LAN 1 uses a Star Topology:** All devices connect to a central switch. This is scalable, easy to maintain, and localizes failures. The switch efficiently manages MAC addresses for fast frame forwarding.
- **LAN 2 uses an Extended Star/Bus-Hub-Like Topology:** Each device connects to an individual switch—more segmentation, can demonstrate cascading switch behavior, and prepares for larger enterprise-like topologies.

- **Router operates as a bridge between LANs:** Segments broadcast domains and enables inter-subnet routing.

6. Why These Steps?

- **Logical separation of networks (subnets)** increases security and manageability.
- **Default gateway assignments** allow for routing outside the local subnet.
- **Switches provide efficient, full-duplex communication in star configurations.**
- **Router manages cross-network communication**, enabling communication between the green and blue zones.

Device	IP Address	Subnet Mask	Default Gateway
PC-PT 0	10.10.0.2	255.0.0.0	10.10.0.1
PC-PT 1	10.10.0.3	255.0.0.0	10.10.0.1
PC-PT 2	10.10.0.4	255.0.0.0	10.10.0.1
PC-PT 3	10.10.0.5	255.0.0.0	10.10.0.1
PC-PT 4	192.168.1.2	255.255.255.0	192.168.1.1
PC-PT 5	192.168.1.3	255.255.255.0	192.168.1.1
PC-PT 6	192.168.1.4	255.255.255.0	192.168.1.1
PC-PT 7	192.168.1.5	255.255.255.0	192.168.1.1
Router G0/1	10.10.0.1	255.0.0.0	—
Router G0/0	192.168.1.1	255.255.255.0	—